Anoushka Alavilli

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Research Interests: My research lies at the intersection of reinforcement learning and model-based control, focusing on **sim-to-real transfer** and data-efficient **adaptation** of dynamics models and control policies. I aim to enable robots to **continually learn and adapt** in real-world environments, building on knowledge and adaptation strategies acquired in simulation.

Website: https://anoushkapalavilli.github.io/portfolio/

Education

Carnegie Mellon University **Pittsburgh, PA**
Ph.D. student at the Robotics Institute, School of Computer Science *August 2024 – current*
Coursework: Deep Reinforcement Learning, Introduction to Robot Learning, Integrated Planning & Learning, Kinematics, Dynamics & Control, and Math Fundamentals for Robotics

Carnegie Mellon University **Pittsburgh, PA**
M.S. in Electrical & Computer Engineering *August 2022 – December 2023*
GPA – 3.89/4.0
Coursework: Optimal Control & Reinforcement Learning, Intro. to Machine Learning for Engineers, Robot Dynamics & Analysis, Applied Stochastic Processes, Computer Vision, Digital Signal Processing.

Tufts University **Medford, MA**
B.S. in Electrical Engineering *September 2018 – May 2022*
GPA – 3.91/4.0, graduated *summa cum laude*
Relevant Coursework: Convex Optimization, Linear Systems, Feedback Control Systems, Probabilistic Systems Analysis, Distributed Machine Learning and Control, Computational Geometry, Mathematical Aspects of Data Analysis, Data Structures.

Experience

Robotics Institute, Carnegie Mellon University **Pittsburgh, PA**
PhD Student, Co-advised by Prof. Jeff Schneider and Prof. Guanya Shi *August 2024 – current*

- Conducting research in **reinforcement learning and model-based control**, focusing on sim-to-real transfer and data-efficient adaptation of dynamics models and control policies, with applications in autonomous off-road driving.
- Developing **model-guided policy adaptation methods** that leverage simulation-pretraining and limited real-world experience to improve control policies during deployment. Early results show these model-guided updates are more stable than policy gradient updates, yielding robust learning with limited data.
- Investigating **vision-based terrain adaptation**, incorporating real-world visual observations to adapt simulation-trained models and policies to previously unseen terrain conditions.
- Developed a hierarchical **RL–MPPI framework for high-speed off-road navigation**, using learned value functions as execution-aware planning costs to integrate reinforcement learning and model predictive control. Achieved **86% navigation success vs. 55% for baseline MPPI** on unseen simulated off-road tasks. Published as a **full paper with oral presentation at IEEE IROS 2026** (see Publications [1]).
- Subsequently integrated and deployed the RL–MPPI framework on a **full-scale Yamaha Viking ATV** for real-world off-road navigation.
- Co-leading a weekly CMU graduate student reading group on dynamics modeling and control.

FieldAI **Irvine, CA**

Robotics Research Intern, Federal Off-road Driving Team *May – August 2026*

- Developed a **physics-structured dynamics model learning** framework for mobile robots in challenging environmental conditions, combining a Lagrangian dynamics prior with a structured learned basis function to adapt to environmental interactions and unknown disturbances.
- Designed a method for simulation-pretraining and data-efficient **online system identification and adaptation**, using closed-form ridge regression to update low-dimensional dynamics parameters from recent robot observations, enabling rapid adaptation to changing environmental conditions.
- Evaluated adaptation to **changing physical conditions and dynamics uncertainty**, including external disturbances, unknown payload masses, and shifting loads. Demonstrated improved multi-step prediction accuracy over baselines in IsaacLab.
- Integrated the learned adaptive dynamics model into a 60 Hz **closed-loop Model Predictive Path Integral (MPPI)** controller.
- **Won best poster award** for this work at IROS 2026 workshop *Bridging Perspectives in Navigation* (see Publications [2]).

NASA Jet Propulsion Laboratory **Pasadena, CA**

Intern, Aerial Mobility Robotics Group (347T) *February – August 2024*

- Performed research in dynamics modeling for off-road driving as part of the controls & planning team on the DARPA RACER project.
- Designed and implemented a **hybrid terradynamics model** for the offroad vehicle. Incorporated DINOv2 visual foundation model (VFM) feature information from the terrain to enable vision-based learned adaptation of the terradynamics model.
- Integrated the dynamics modeling and visual feature mapping into the team's **ROS autonomy stack**. Wrote all code using Python, C++, and ROS.
- Published and presented resulting research at **IEEE ICRA 2025** (see Publications [3]).

Robotic Exploration Lab, Carnegie Mellon University **Pittsburgh, PA**

Master's student researcher under Professor Zachary Manchester

TinyMPC *January – December 2023*

- Developed a fast and memory-efficient **model predictive control (MPC) solver for embedded applications** with trajectory tracking and real-time obstacle avoidance. Showcased functionality on a Crazyflie drone.
- Formulated and implemented the Alternating Direct Method of Multipliers (ADMM) to solve the underlying MPC problem.
- Published this work as a conference paper to **IEEE ICRA 2024** within a student group (see Publications [4]) and won award for **Best Paper in Automation**.

NASA Jet Propulsion Laboratory **Pasadena, CA**

Summer Intern, Mars 2020 Instrument Data Systems Operations *May – July 2023*

- Designed a variation of Gaussian mixture models with probabilistic uncertainty quantification (exceeding initial assignment) to predict data pipeline outputs for Mars 2020 mission downlink operations. Model circumvents the prior dependence on image content, instead only using instrument metadata from Mars, to make output predictions.

Publications

[1] **Alavilli, A.**, Song, R., Shi, G., & Schneider, J. (2026). Arxiv link. Value Function-Guided MPPI for Hierarchical Planning and Control in Off-Road Autonomous Navigation (*Published as full paper with oral presentation* at IEEE IROS 2026.)
[2] **Alavilli, A.**, Noren, C., Gibson, J., Shi, G., Schneider, J., & Khattak, S. (2026). Model Learning and Online Adaptation for Environmental Disturbances in Autonomous Off-Road Driving. Paper link. (*Accepted as a poster* to the IEEE IROS 2026 Workshop on Bridging Perspectives in Navigation.)
[3] Gibson, J.*, **Alavilli, A.***, Tevere, E., Evangelos, T., & Spieler, P. (2024). Dynamics Modeling using Visual Terrain Features for High-Speed Autonomous Off-Road Driving. Arxiv link. (Published in IEEE ICRA 2025 Conference Proceedings.) * These authors contributed equally
[4] Nguyen, K.*, Schoedel, S.*, **Alavilli, A.***, Plancher, B., & Manchester, Z. (2023). TinyMPC: Model-Predictive Control on Resource-Constrained Microcontrollers. arXiv preprint arXiv:2310.16985. Arxiv link. (Published in IEEE ICRA 2024 Conference Proceedings.) * These authors contributed equally.
[5] **Alavilli, A.**, & Vu, M. (2022). PLAN: A Leafcutter Ant Colony Optimization Algorithm for Ride-Hailing Services. Publication link. (Published in ACM Genetic and Evolutionary Computation Conference (GECCO) 2022 Proceedings.)

Awards

- **IROS 2026 Bridge4Navigation Workshop Best Poster Award** for *Model Learning and Online Adaptation for Environmental Disturbances in Autonomous Off-Road Driving* (see Publications [2]).
- **ICRA 2024 Best Paper Award in Automation** for TinyMPC (see Publications [4]).
- **Harry Poole Burden Prize in Electrical Engineering**, Tufts University, Spring 2022 (Awarded for the best design or research project done by electrical engineering undergraduate students).
- **Morris and Sid Heyman Prize Scholarship**, Tufts University, Spring 2021 (Awarded for academic achievement and future promise).

Talks and Presentations

- **First-author paper presentation** at IROS 2026 for *Value Function-Guided MPPI for Hierarchical Planning and Control inOff-Road Autonomous Navigation*.
- **First-author paper presentation** at ICRA 2025 for *Dynamics Modeling using Visual Terrain Features for High-Speed Autonomous Off-Road Driving*.
- **Guest lecture** on autonomous off-road driving and MPPI for the course *16-745 Optimal Control and Reinforcement Learning* taught by Professor Chris Atkenson at Carnegie Mellon University in Spring 2026.
- **Guest lecture** on non-technical introduction to research in autonomous off-road driving for the course *16-199 Robotics in the Real World* taught by Professor Aaron Steinfeld at Carnegie Mellon University in Fall 2026.
- **Guest lecture** on convolutional neural networks for the course *Fundamentals and Engineering Applications of AI* taught by Professor Ozan Tonguz at Carnegie Mellon University in Spring 2023.
- **First-author paper presentation** at ACM GECCO 2022 conference for undergraduate research.

Skills

Programming languages: Python, Julia, MATLAB, C++

Computational tools: Linux, ROS2, Docker, IsaacLab, JAX, PyTorch, MeshCat, CVX/CONVEX.jl, Git, Codex, Claude Code, LaTeX,.